

Day 3



Aggregation functions

✓ **Aggregate values in multiple rows to one value**

Data Output		Explain
	amount numeric (5,2) 🔒	
1	1.99	
2	0.99	
3	6.99	
4	0.99	
5	4.99	
6	2.99	

SUM



Data Output		Expl
	sum numeric 🔒	
1	67416.51	

Aggregation functions

✓ **Aggregate values in multiple rows to one value**

Data Output		Explain
	 amount numeric (5,2) 	
1	1.99	
2	0.99	
3	6.99	
4	0.99	
5	4.99	
6	2.99	

AVG



Data Output		Explain
	 avg numeric 	
1	4.20	



Most common aggregation functions

SUM()
AVG()
MIN()
MAX()
COUNT()



SYNTAX

```
SELECT  
SUM(amount)  
FROM payment
```

Data Output		Expl
	sum numeric 	
1	67416.51	

SYNTAX

```
SELECT  
COUNT (*)  
FROM payment
```

What we can't do...

```
SELECT  
SUM(amount)  
payment_id  
FROM payment
```

Data Output		Expl
	sum numeric 	
1	67416.51	

No mixing possible!

```
SELECT  
SUM(amount)  
payment_id ←  
FROM payment
```

Only possible with
grouping!

Data Output		Expl
	sum	
	numeric 	
1	67416.51	

Multiple aggregations is possible!

```
SELECT  
SUM(amount) ,  
COUNT(*) ,  
AVG(amount)  
FROM payment
```

	Data Output	Explain	Messages
	sum numeric	count bigint	avg numeric
1	67416.51	16049	4.20



Challenge

Challenge

Your manager wants to which of the two employees (staff_id) is responsible for more payments?

Which of the two is responsible for a higher overall payment amount?

How do these amounts change if we don't consider amounts equal to 0?

Write two SQL queries to get the answers!

Result

	Data Output	Explain	Messages
	 staff_id smallint	sum numeric	count bigint
1	2	33927.04	7992
2	1	33489.47	8057

	Data Output	Explain	Messages
	 staff_id smallint	sum numeric	count bigint
1	2	33927.04	7983
2	1	33489.47	8042

GROUP BY

✓ Used to **GROUP** aggregations **BY** specific columns

	Data Output	Explain	Messages	No
	customer_id smallint	amount numeric (5,2)		
1	269	1.99		
2	269	0.99		
3	269	6.99		
4	269	0.99		
5	269	4.99		
6	269	2.99		
7	270	1.99		
8	270	4.99		

SUM



	Data Output	Explain	Messa
	customer_id smallint	sum numeric	
1	1	118.68	
2	2	128.73	
3	3	135.74	
4	4	81.78	
5	5	144.62	

SYNTAX

```
SELECT  
customer_id,  
SUM(amount)  
FROM payment  
GROUP BY customer_id
```

	Data Output	Explain	Message
	customer_id smallint	sum numeric	
1	1	118.68	
2	2	128.73	
3	3	135.74	
4	4	81.78	
5	5	144.62	

SYNTAX

```
SELECT  
customer_id,  
SUM(amount)  
FROM payment  
WHERE customer_id >3  
GROUP BY customer_id
```

	Data Output	Explain	Message
	customer_id smallint 🔒	sum numeric 🔒	
1	1	118.68	
2	2	128.73	
3	3	135.74	
4	4	81.78	
5	5	144.62	

SYNTAX

```
SELECT  
customer_id,  
SUM(amount)  
FROM payment  
WHERE customer_id >3  
GROUP BY customer_id  
ORDER BY customer_id
```

	Data Output	Explain	Message
	customer_id smallint	sum numeric	
1	1	118.68	
2	2	128.73	
3	3	135.74	
4	4	81.78	
5	5	144.62	

SYNTAX

```
SELECT  
customer_id,  
SUM(amount)  
FROM payment  
GROUP BY customer_id
```

Every column:
In GROUP BY or
in aggregate functions

SYNTAX

```
SELECT  
customer_id,  
SUM(amount) }  
FROM payment  
GROUP BY customer_id
```

Every column:
In GROUP BY or
in aggregate functions



Challenge

Challenge

There are two competitions between the two employees.

Which employee had the highest sales amount in a single day?

Which employee had the most sales in a single day (not counting payments with amount = 0)?

Write two SQL queries to get the answers!

Result

	Data Output	Explain	Messages	Notification
	 date	 staff_id smallint	 sum numeric	 count bigint
1	2020-04-30	2	2866.42	658
2	2020-04-30	1	2736.75	625
3	2020-03-21	2	1505.52	348



Challenge

Challenge

Your manager wants to get a better understanding of the films.

That's why you are asked to write a query to see the

- Minimum
- Maximum
- Average (rounded)
- Sum

of the replacement cost of the films.

Write a SQL query to get the answers!

Result

	Data Output	Explain	Messages	Notifications
	 min numeric	 max numeric	 avg numeric	 sum numeric
1	9.99	29.99	19.98	19984.00

Solution

```
SELECT  
MIN(replacement_cost),  
MAX(replacement_cost),  
ROUND(AVG(replacement_cost),2) AS AVG,  
SUM(replacement_cost)  
FROM film
```

HAVING

✓ Used to **FILTER** Groupings BY aggregations

Data Output		Explain	Messages	Notifica
	staff_id smallint	date date	sum numeric	count bigint
1	2	2020-04-30	2866.42	658
2	1	2020-04-30	2736.75	625
3	2	2020-03-21	1505.52	348
4	1	2020-03-01	143	
5	2	2020-03-19	140	

HAVING
COUNT(*)>400



Data Output		Explain	Messages	Notifica
	staff_id smallint	date date	sum numeric	count bigint
1	2	2020-04-30	2866.42	658
2	1	2020-04-30	2736.75	625

Note!
HAVING can only be used
with GROUP BY!

SYNTAX

```
SELECT  
customer_id,  
SUM(amount)  
FROM payment  
GROUP BY customer_id  
HAVING SUM(amount) > 200
```

	Data Output	Explain	Mes
	customer_id smallint	sum numeric	
1	526	221.55	
2	148	216.54	

Solution

```
SELECT  
MIN(replacement_cost),  
MAX(replacement_cost),  
ROUND(AVG(replacement_cost),2) AS AVG,  
SUM(replacement_cost)  
FROM film
```



Challenge

Challenge

In 2020, April 28, 29 and 30 were days with very high revenue. That's why we want to focus in this task only on these days (filter accordingly).

Find out what is the **average payment amount grouped by customer and day** – consider only the days/customers with more than 1 payment (per customer and day).
Order by the average amount in a descending order.

Write a SQL query to find out!

Result

	Data Output	Explain	Messages	Notifications
	customer_id smallint	date date	avg_amount numeric	count bigint
1	459	2020-04-29	10.49	2
2	443	2020-04-28	9.49	2
3	510	2020-04-28	9.49	2